CHAPTER - 2

Expression of Concentration

Chapter Overview

In this chapter, students will learn:

- Methods of expressing concentration of solutions
- Normality, molarity, molality, and formality
- Percentage solutions
- Parts per million (ppm) and parts per billion (ppb)
- Preparation of standard solutions

2.1 Methods of Expressing Strength of Solutions

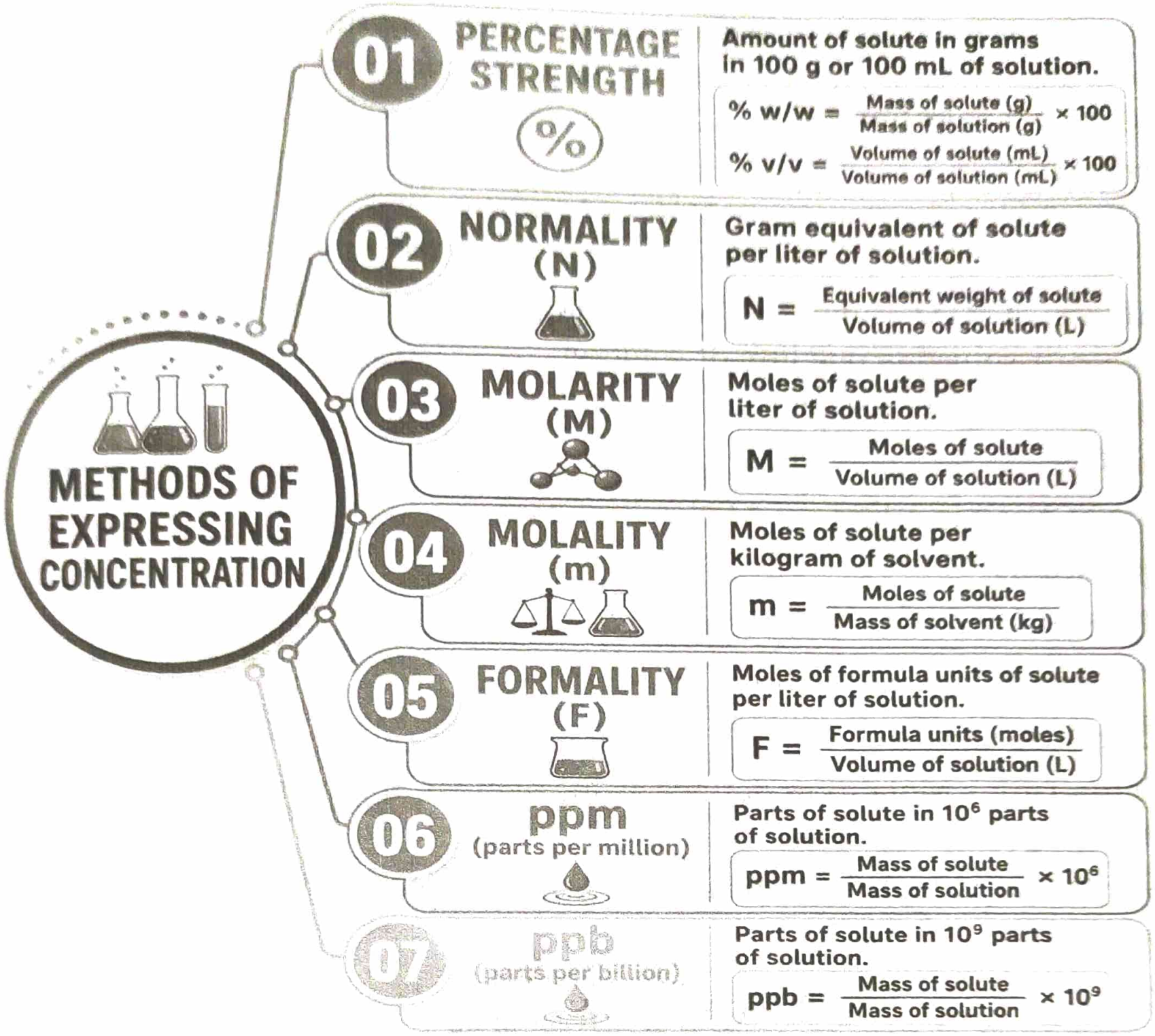
The strength or concentration of a solution indicates the amount of solute dissolved in a given quantity of solvent or solution. In pharmaceutical analysis, concentration of solutions is very important for accurate preparation of medicines, reagents, and analytical solutions.

Different methods are used to express the concentration of solutions depending on the nature of the substance and purpose of analysis.

Why Concentration is Important?

Correct concentration helps in:

- Accurate drug preparation
- Proper dosage
- Reliable analytical results
- Quality control of medicines



2.2 Normality, Molarity, Molality and Formality

1. **Normality (N):** Normality is defined as the number of gram equivalents of solute present in one litre of solution. It is commonly used in acid–base reactions and redox titrations.

Formula:

$$N = \frac{Gram\ Equivalents\ of\ Solute}{Volume\ of\ Solution\ in\ Litres}$$

Example: - A 1 N hydrochloric acid solution contains one gram equivalent of HCl dissolved in one litre of solution.

$$HCl + NaOH \rightarrow NaCl + H_2O$$

Applications

- Acid–base titrations
- Redox analysis
- Pharmaceutical assays

2. **Molarity (M):** Molarity is defined as the number of moles of solute present in one litre of solution. It is one of the most commonly used methods for expressing concentration.

Formula:

$$M = \frac{\textit{Moles of Solute}}{\textit{Volume of Solution in Litres}}$$

Example: A 1 M sodium chloride solution contains one mole of sodium chloride dissolved in one litre of solution.

Molecular weight of NaCl = 58.5 g/mol

Therefore,

58.5 g of NaCl in 1 litre = 1 M NaCl solution

Applications

- Preparation of laboratory solutions
- Pharmaceutical calculations
- Chemical analysis

3. **Molality (m):** - Molality is defined as the number of moles of solute present in one kilogram of solvent.

Unlike molarity, molality depends on mass and not volume.

Formula:

$$m = \frac{\textbf{Moles of Solute}}{\textbf{Mass of Solvent in kg}}$$

Example: One mole of glucose dissolved in one kilogram of water forms a 1 molal solution.

Applications

- Physical chemistry studies
- Determination of colligative properties

4. **Formality (F):** Formality is defined as the number of gram formula weights of solute present in one litre of solution.

It is mainly used for ionic compounds that do not exist as molecules in solution.

Formula:

$$F = \frac{\textbf{Gram Formula Weight of Solute}}{\textbf{Volume of Solution in Litres}}$$

Example: A solution containing one gram formula weight of sodium chloride in one litre is called a 1 F solution.

Applications

- Ionic compound solutions
- Analytical chemistry calculations

Comparison Between Normality, Molarity, Molality and Formality

Basis	Normality	Molarity	Molality	Formality
Definition	Gram equivalents per litre	Moles per litre	Moles per kg of solvent	Formula weights per litre
Unit	N	M	m	F
Depends On	Equivalent weight	Molecular weight	Mass of solvent	Formula weight
Volume Dependent	Yes	Yes	No	Yes
Common Use	Titrations	General solutions	Physical chemistry	Ionic compounds

2.3 Percentage Solutions

Percentage solutions express the amount of solute present in 100 parts of solution. It is one of the simplest methods of expressing concentration.

1. **Percentage Weight by Weight (% w/w)**

   It represents the number of grams of solute present in 100 g of solution.

   Formula:

$$\% \, w/w = \frac{\textbf{Weight of Solute}}{\textbf{Weight of Solution}} \times \mathbf{100}$$

   Example: 10 g of sodium chloride in 100 g solution = 10% w/w solution.

2. **Percentage Weight by Volume (% w/v)**

   It represents the number of grams of solute present in 100 mL of solution.

   Formula:

$$\% \, w/v = \frac{\textbf{Weight of Solute}}{\textbf{Volume of Solution}} \times \mathbf{100}$$

   Example: 5 g glucose in 100 mL solution = 5% w/v solution.

3. **Percentage Volume by Volume (% v/v)**

   It represents the number of millilitres of solute present in 100 mL of solution.

   Formula:

$$\% \, v/v = \frac{\textbf{Volume of Solute}}{\textbf{Volume of Solution}} \times \mathbf{100}$$

   Example: 70 mL alcohol in 100 mL solution = 70% v/v solution.

2.4 ppm and ppb

ppm (Parts Per Million): ppm is used to express very dilute concentrations of substances. It represents one part of solute present in one million parts of solution.

Formula:

$$ppm = \frac{\text{Amount of Solute}}{\text{Amount of Solution}} \times 10^6$$

Example: 1 mg of substance in 1 litre of water ≈ 1 ppm.

ppb (Parts Per Billion): ppb is used for extremely dilute solutions. It represents one part of solute present in one billion parts of solution.

Formula:

$$ppb = \frac{\text{Amount of Solute}}{\text{Amount of Solution}} \times 10^9$$

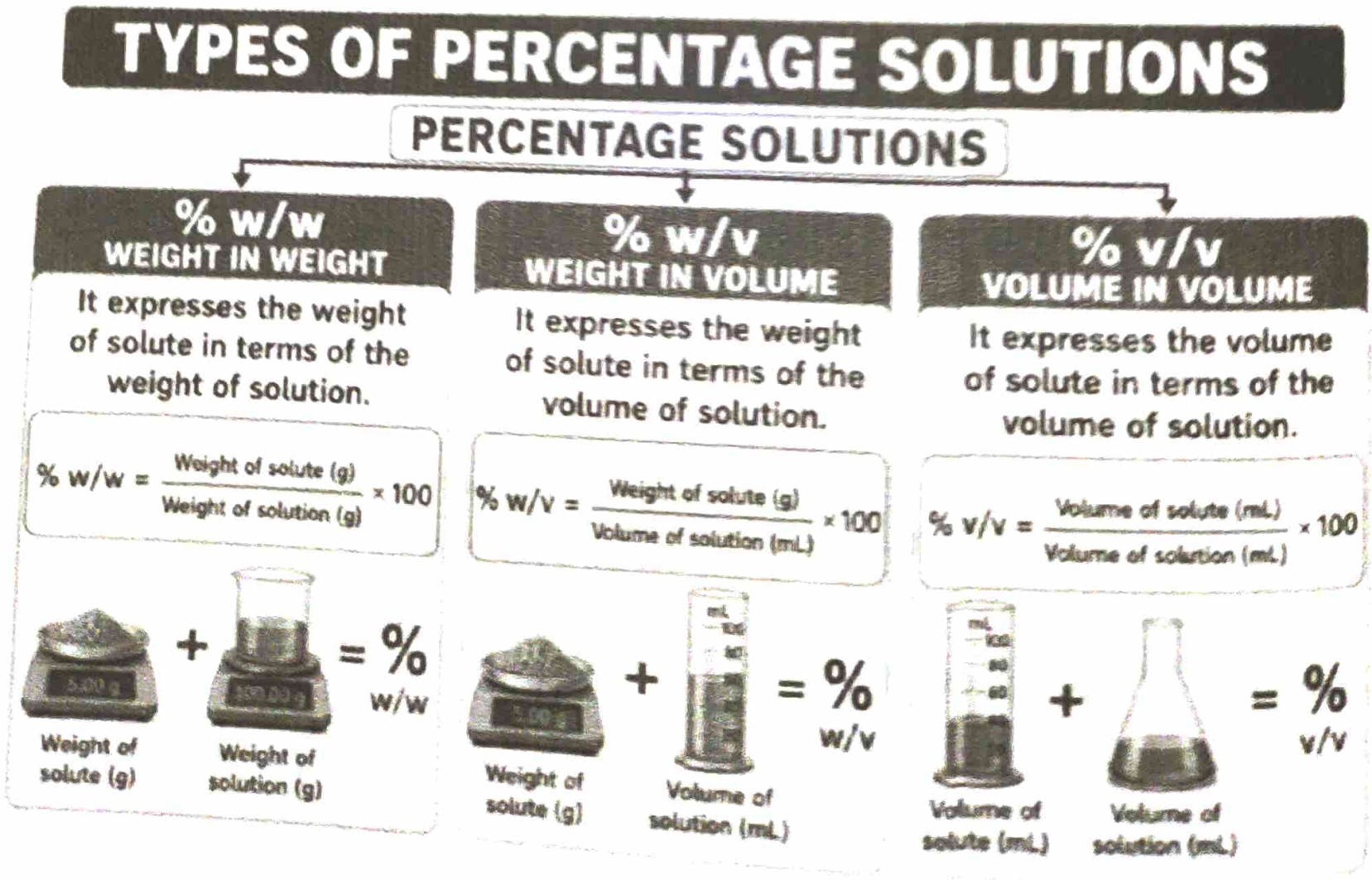
Example: 1 μg of substance in 1 litre of water ≈ 1 ppb.

Applications of ppm and ppb

- Detection of impurities
- Environmental analysis
- Pharmaceutical quality control
- Trace element analysis

2.5 Preparation of Standard Solutions

A standard solution is a solution whose concentration is accurately known. Standard solutions are widely used in pharmaceutical analysis and titration methods.



Comparison Between Normality, Molarity, Molality and Formality

Basis	Normality	Molarity	Molality	Formality
Definition	Gram equivalents per litre	Moles per litre	Moles per kg of solvent	Formula weights per litre
Unit	N	M	m	F
Depends On	Equivalent weight	Molecular weight	Mass of solvent	Formula weight
Volume Dependent	Yes	Yes	No	Yes
Common Use	Titrations	General solutions	Physical chemistry	Ionic compounds

2.3 Percentage Solutions

Percentage solutions express the amount of solute present in 100 parts of solution. It is one of the simplest methods of expressing concentration.

1. **Percentage Weight by Weight (% w/w)**

   It represents the number of grams of solute present in 100 g of solution.

   Formula:

$$\% \, w/w = \frac{\textbf{Weight of Solute}}{\textbf{Weight of Solution}} \times \mathbf{100}$$

   Example: 10 g of sodium chloride in 100 g solution = 10% w/w solution.

2. **Percentage Weight by Volume (% w/v)**

   It represents the number of grams of solute present in 100 mL of solution.

   Formula:

$$\% \, w/v = \frac{\textbf{Weight of Solute}}{\textbf{Volume of Solution}} \times \mathbf{100}$$

   Example: 5 g glucose in 100 mL solution = 5% w/v solution.

3. **Percentage Volume by Volume (% v/v)**

   It represents the number of millilitres of solute present in 100 mL of solution.

   Formula:

$$\% \, v/v = \frac{\textbf{Volume of Solute}}{\textbf{Volume of Solution}} \times \mathbf{100}$$

   Example: 70 mL alcohol in 100 mL solution = 70% v/v solution.

2.4 ppm and ppb

ppm (Parts Per Million): ppm is used to express very dilute concentrations of substances. It represents one part of solute present in one million parts of solution.

Formula:

$$ppm = \frac{\text{Amount of Solute}}{\text{Amount of Solution}} \times 10^6$$

Example: 1 mg of substance in 1 litre of water ≈ 1 ppm.

ppb (Parts Per Billion): ppb is used for extremely dilute solutions. It represents one part of solute present in one billion parts of solution.

Formula:

$$ppb = \frac{\text{Amount of Solute}}{\text{Amount of Solution}} \times 10^9$$

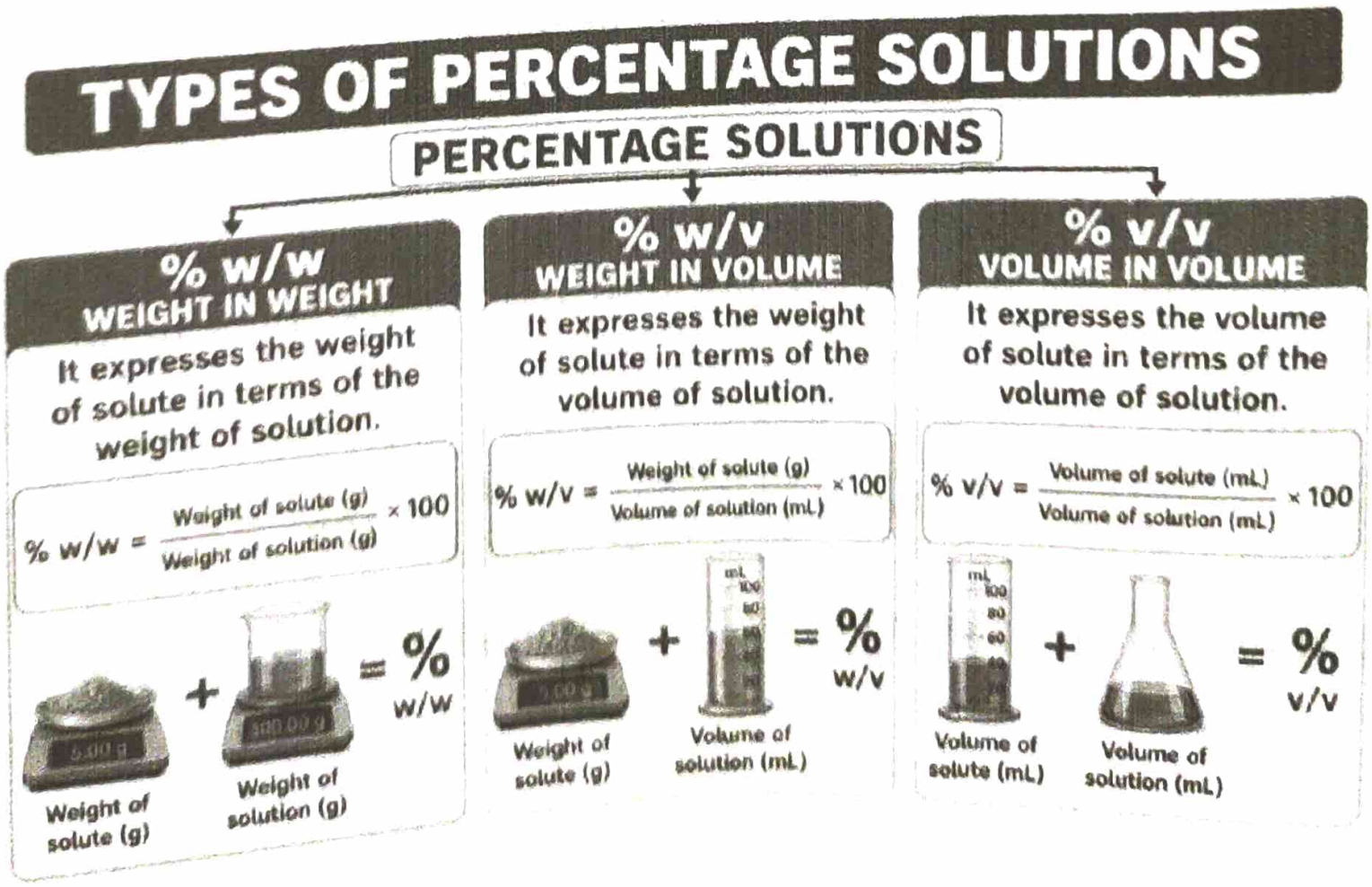
Example: 1 μg of substance in 1 litre of water ≈ 1 ppb.

Applications of ppm and ppb

- Detection of impurities
- Environmental analysis
- Pharmaceutical quality control
- Trace element analysis

2.5 Preparation of Standard Solutions

A standard solution is a solution whose concentration is accurately known. Standard solutions are widely used in pharmaceutical analysis and titration methods.



a. Primary Standard Solution

A primary standard solution is prepared using a highly pure, stable, and non-hygroscopic substance.

Example

- Sodium carbonate
- Oxalic acid

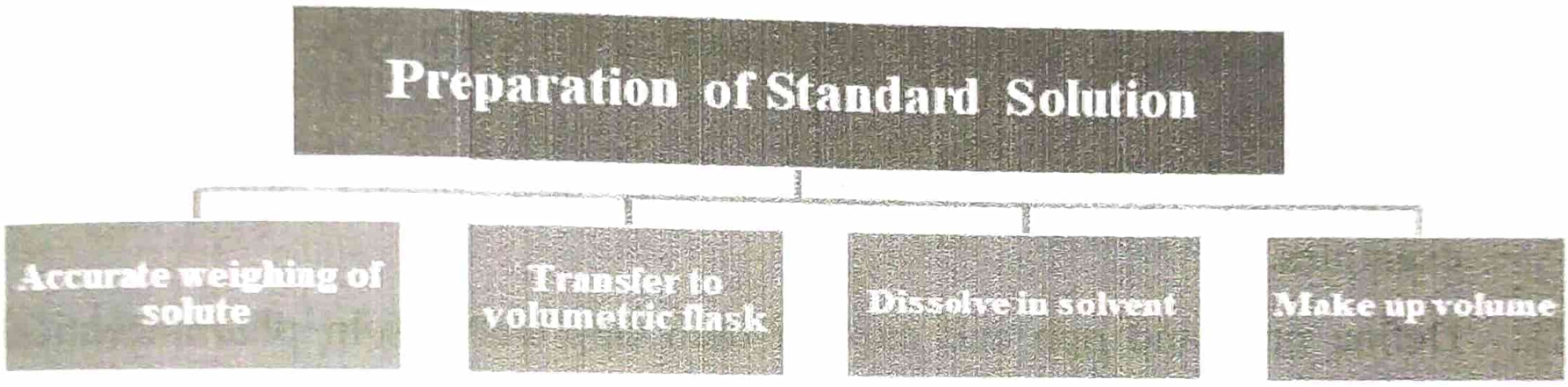
b. Secondary Standard Solution

A secondary standard solution is a solution whose concentration is determined using a primary standard solution.

Example

- Hydrochloric acid
- Sodium hydroxide

Steps in Preparation of Standard Solutions



Applications in Pharmacy

- Drug assay
- Quantitative analysis
- Quality control testing
- Preparation of reagents

Key Points to Remember

Normality = Equivalent/Litre

Molarity = Mole/Litre

Molality = Mole/kg solvent

ppm is used for dilute solutions

Standard solutions are essential in titrations

Question Bank

1. Define concentration of a solution and explain its importance in pharmaceutical analysis.
2. Discuss the different methods used for expressing the strength of solutions.
3. Define normality and explain its applications in pharmaceutical analysis.
4. Define molarity and describe its significance in preparation of pharmaceutical solutions.
5. Differentiate between normality and molarity.
6. Explain molality with suitable examples and applications.
7. Define formality and discuss its importance in analytical chemistry.
8. Compare normality, molarity, molality, and formality.
9. Explain percentage solutions and describe different types of percentage expressions.
10. Differentiate between % w/w, % w/v, and % v/v solutions with examples.
11. Define ppm and ppb and explain their applications in pharmaceutical analysis.
12. Differentiate between ppm and ppb.
13. What is a standard solution? Explain its importance in pharmaceutical analysis.
14. Differentiate between primary standard and secondary standard solutions with suitable examples.
15. Describe the preparation of a standard solution with the help of a flowchart or suitable diagram.

Multiple Choice Questions

1. The concentration expressed as the number of moles of solute present in one litre of solution is called:

a) Normality
b) Molality
c) Molarity
d) Formality

2. Which concentration unit is independent of temperature?

a) Molarity
b) Normality
c) Formality
d) Molality

3. Normality is defined as the number of:

a) Moles per litre of solution
b) Gram equivalents per litre of solution
c) Moles per kilogram of solvent
d) Formula weights per litre of solution

4. A solution containing 5 g of glucose in 100 mL of solution is expressed as:

a) 5% w/w
b) 5% v/v
c) 5% w/v
d) 0.5% w/v

5. Which of the following is commonly used to express very dilute concentrations?

a) Molarity
b) Formality
c) ppm
d) Normality

6. One ppm is approximately equal to:

a) 1 g/L
b) 1 mg/L
c) 1 µg/L
d) 1 ng/L

7. Which of the following is a primary standard substance?

a) Hydrochloric acid
b) Sodium hydroxide
c) Oxalic acid
d) Ammonium hydroxide

8. A standard solution is a solution whose:

a) Colour is known
b) pH is known
c) Volume is fixed
d) Concentration is accurately known

College of Pl
LIBRARY, V
Acc. No.
Call No.

9. Formality is mainly used for:

a) Organic compounds only
b) Ionic compounds
c) Gaseous mixtures
d) Colloidal solutions